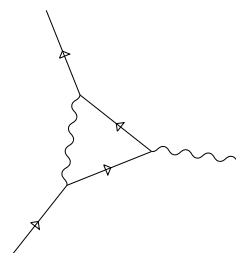




quantum
the ideas of $\wedge$ field theory



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THE IDEAS OF QUANTUM FIELD THEORY

(in progress)

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ABSTRACT. The goal of this article is to communicate some flavor of Quantum Field theory to younger audience via demonstrating certain foundational calculations. This is an article that I would have liked to read back in my middle year of undergraduation.

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1. INTRODUCTION

One of the most profound ideas in QFT is that, particles are seen as quantized fields, and field are seen as represenations of Lorentz group. In the words of Wigner,

Particles are irreducible representations of the Poincare group.

But before we begin to digest this beauty, we must confront the challenges amidst which this theory was born as a beacon of hope. Quantum field theory was introduced as a new paradigm in physics in the 20th century. It is a **multi-particle, local, quantum mechanical** and a **Lorentz invariant microscopic** theory of physics. The origins of many of these issues lie in reconciling special relativity and quantum mechanics together.

1.1. The range of physics that QFT can explain.

1.2. Special relativity $\times$ Quantum mechanics.

$$E = mc^2 \quad \longleftrightarrow \quad \Delta E \geq \frac{\hbar c}{L}$$

Combining mass-energy equivalence and uncertainty principle means that, whenever and wherever the fluctuations in energy are high enough there is a real possibility of particles getting created and annihilated in vacuum. That is, the number of particles need not be conserved in a conventional sense. This is a stark difference from the quantum mechanics we are used to! The use of familiar 1-particle or any fixed particle Schrodinger equation only lead to inconsistent results. For example, we get acausal propagators and an infinite tower of negative energy states. We will describe these failures shortly by indeed brute forcing relativity directly into the fixed particle Quantum Mechanics.

More succinctly, the particles and antiparticles begin to pop up at roughly the length scale of Compton wavelength, $\lambda = \frac{\hbar}{mc}$. The single-particle theory breaks down below this length scale. Just like how the particle nature itself starts to wind down at de Broglie wavelength allowing for a wavelike behavior.

These aspects of *particle creations*, sometimes practically *virtual* due their short lifespan ($\Delta E \sim \frac{1}{\Delta t}$) and the *existence of anti-particles* is predicted quite formally by quantum field theory. Especially through the perturbation theory we see that such a process can be seemingly endless. This usually gives us asymptotic results in QFT, even if they might not converge. We now discuss some important limitations of trying to combine relativity to quantum mechanics/fixed-particle mechanics.

PROBLEM

Acausal propagators.

One usually doesn't care about this in a Quantum Mechanics course, but in the context of relativistic quantum mechanics, the propagators in QM suggest a spacelike propagation of information. Replacing the usual energies with the relativistic ones doesn't seem to make the problem go away either.

For example, consider the amplitude for a free particle to propagate from x_0 to x :

$$U(t) = \langle x | e^{-iHt} | x_0 \rangle$$

with a non-relativistic energy $E = p^2/2m$ would give us,

$$U(t) \sim e^{im(x-x_0)^2/2t}$$

which is non-zero for all x, t ! Changing E to $\sqrt{p^2 + m^2}$ still gives us a non-zero behavior outside the light-cone,

$$U(t) \sim e^{-m\sqrt{x^2-t^2}}.$$

This problem infact doesn't go away immediately even in QFT, but the existence of *anti-matter* cancels out any non-zero amplitude due to the *matter*. Infact this is one of the ways to theoretically understand the existence of anti-matter, to preserve causality in Quantum Mechanics. The principles of causality and locality are quite interconnected in relativity. It's precisely the non-local nature of the above propagator that creates the eventual breakdown in causality. This is why efforts to reconcile Relativity and Quantum mechanics using a Hamiltonian like,

$$H = \sqrt{p^2 + m^2} = m \left(1 + \frac{p^2}{m^2} \right)^{1/2} = m + \frac{p^2}{2m} - \frac{p^4}{8m^3} + \frac{p^6}{16m^5} + \dots$$

fail.

PROBLEM

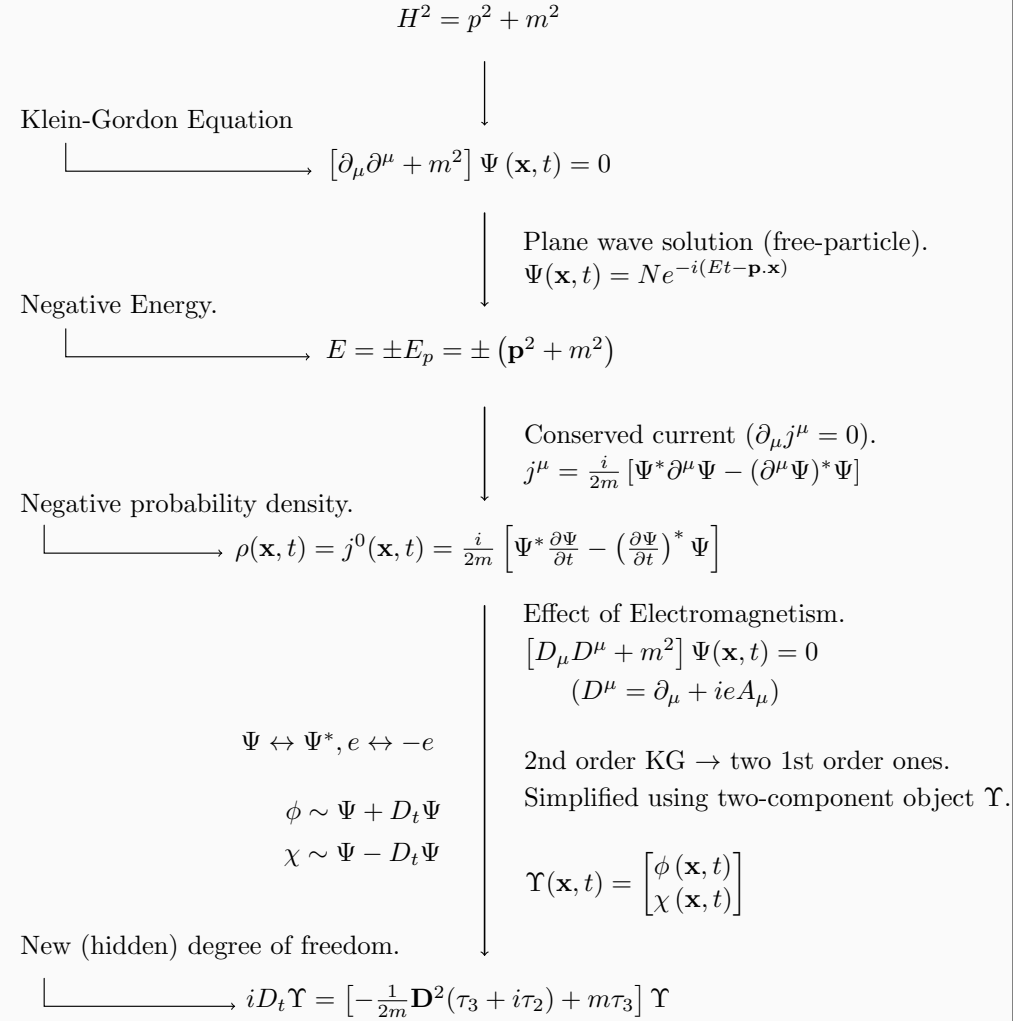
Infinite negative Energy states.

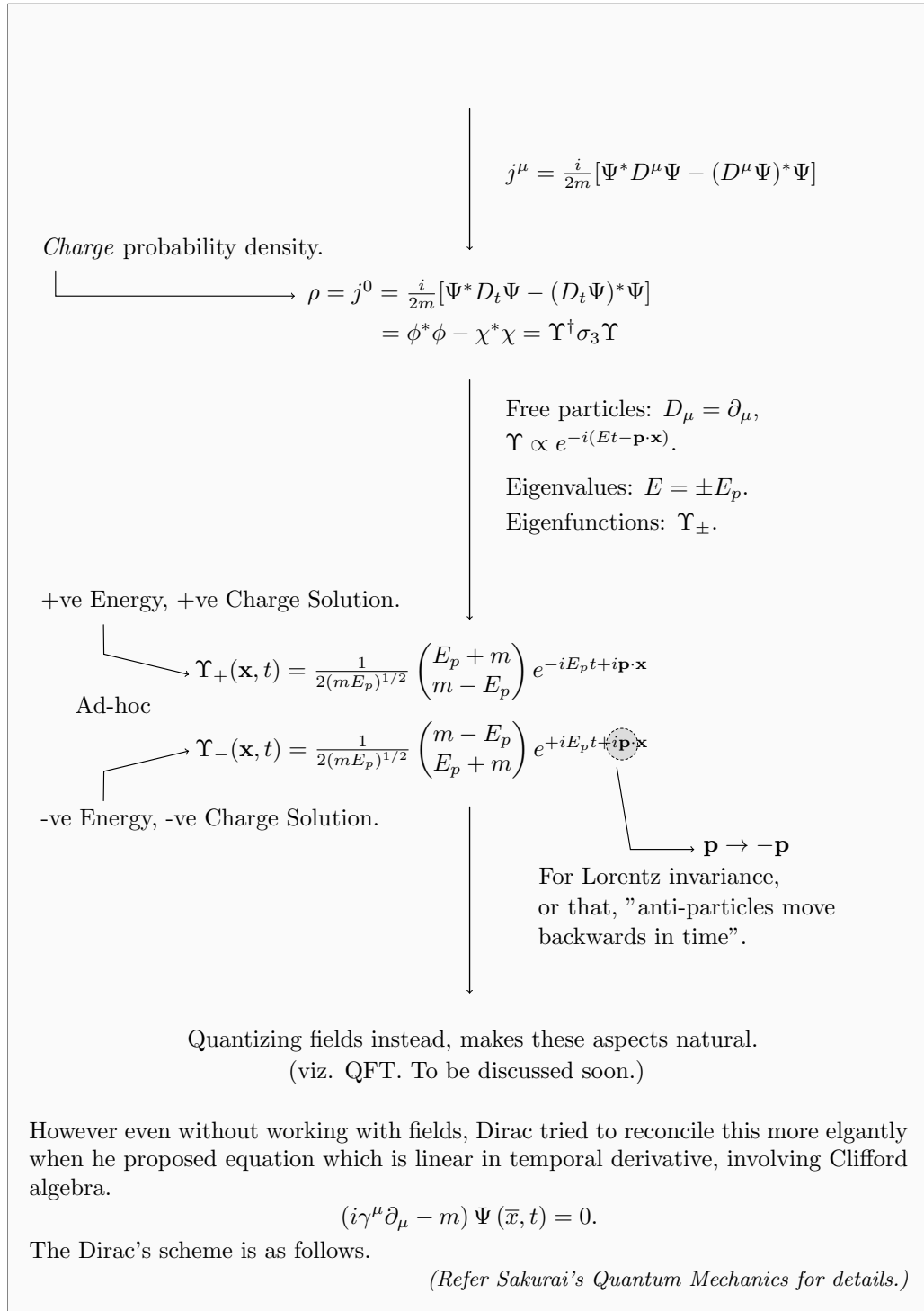
Instead, one can approach using the square of the Hamiltonian, $H^2 = p^2 + m^2$ to modify the Schrodinger equation as:

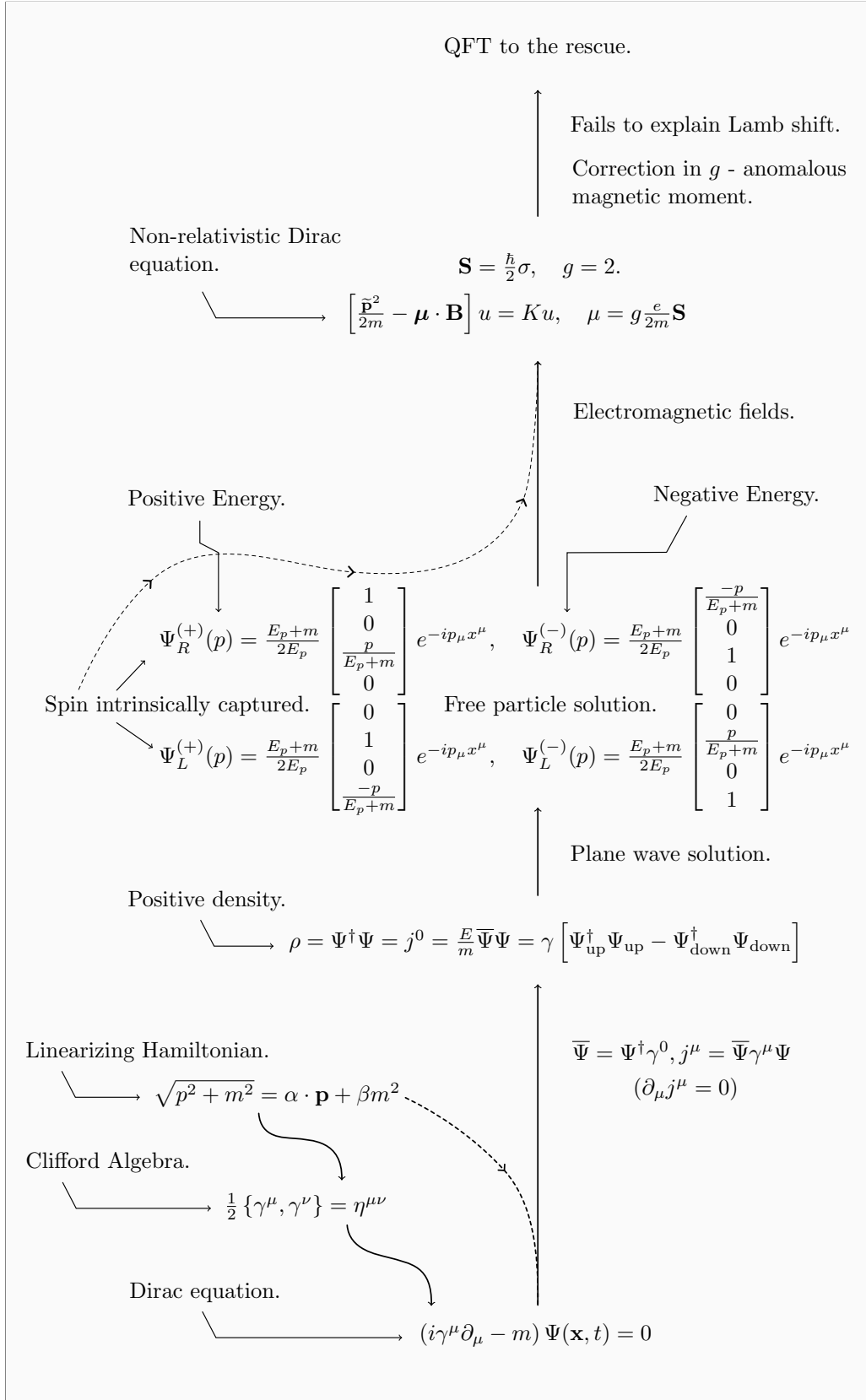
$$\frac{\partial^2}{\partial t^2} |\psi(t)\rangle = H^2 |\psi(t)\rangle$$

$$\left[\frac{\partial^2}{\partial t^2} - \nabla^2 + m^2 \right] \Psi(\bar{x}, t) = 0.$$

This is called the Klein-Gordon equation. But it leads immediately to the possibility of negative energy states again along with negative probability densities. The Klein Gordon attempt to brute force relativity into quantum mechanics takes the following (ad-hoc) route to interpret the above issues.







In 1926, when Schrodinger published his equation explaining the Bohr's energy levels of Hydrogen atom, he had also stumbled upon the Klein-Gordon equation. But he had ignored it since the fine structure (which required spin) was still not possible to explain. The Klein-Gordon efforts to interpret the results of this equation followed. But it was Dirac in early 1928 who successfully managed to linearize the Hamiltonian, inturn also including spin of charged particles naturally. In 1931 Dirac revisited the discussion and proposed the existence of anti-matter to interpret the negative energies. In 1933, such matter was indeed discovered by Carl Anderson. This still gives only a tiny closure and there are issues like Lamb shift and acausal propagators remaining to be solved.

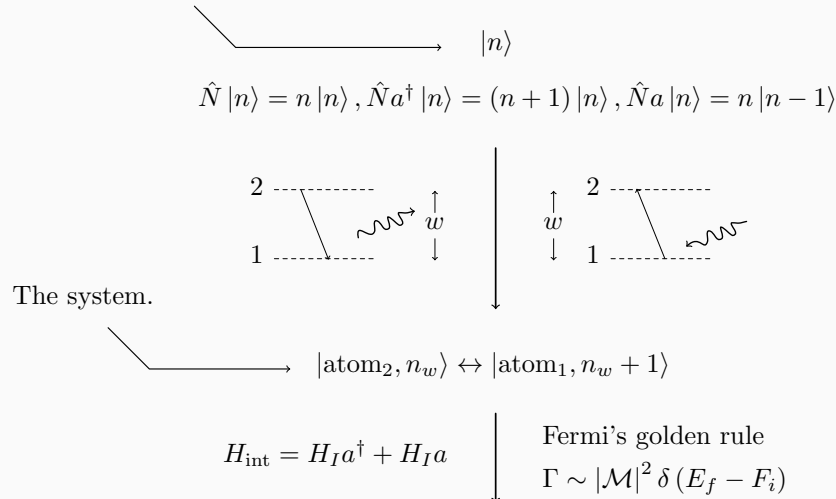
(A good reference for the above history is Wikipedia's article on Dirac equation.)

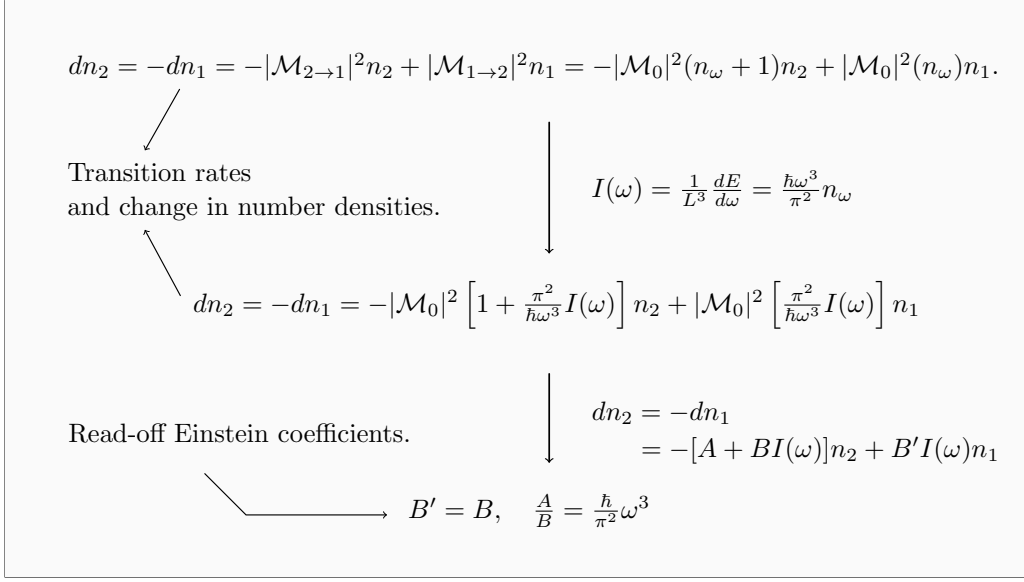
IDEA

Dirac's Blackbody treatment.

In 1927, Dirac had also approached the blackbody radiation in his paper on *The quantum theory of the emission and absorption of radiation*. At that time, two famous discussions led by Planck and Einstein (1916) were famous, describing the blackbody with quantum of light, thermal equilibrium, gas equilibrium and einstein coefficients of atoms related to emission and absoprtion. Dirac then initiated a novel multi-particle approach to quantum-mechanics (as opposed to a fixed-particle approach) to microscopically derive the atomic Einstein coefficients from the first principles without assuming. This is often credited to intiate the idea of canonical quantization and quantum field theory! The multi-particle approach manifests as treating different frequencies as different particles and as exciations to a quantum harmonic oscillator! The equilibrium is then established via creation and annihilation of these particules, which is made explicit by this approach which continues to extend to QFT. Here are some key steps.

Light quanta: $\Delta, 2\Delta, 3\Delta, \dots$
as harmonic oscillators.





IDEA

Fields for a local theory.

As we had already noted, locality is almost a prerequisite for causality. We replace wavefunctions with fields, and these will be our dynamical variables. Following this field approach and using Dirac's harmonic oscillator type quantization (canonical quantization) fixes many of our above issues giving causal propagators and natural occurrence of anti-matter. As we will see in the next section, this quantization involves inheriting the creation and annihilation algebra of harmonic oscillator into the fourier modes of classical fields, which also satisfy the harmonic oscillator equation of motion!

IDEA

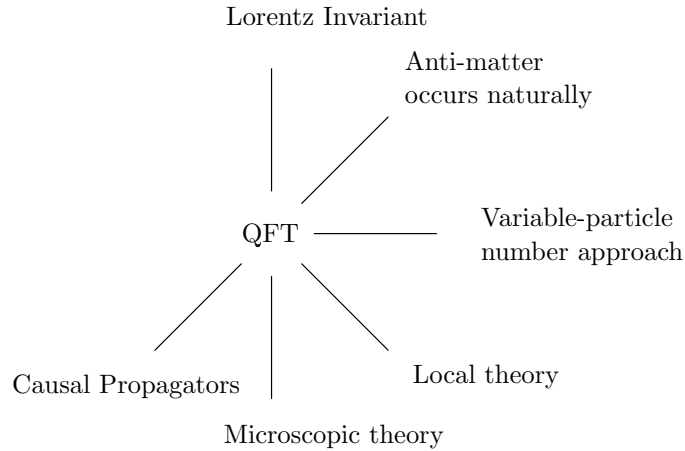
Lorentz Invariance – Space and Time on an equal footing.

Finally we want all our theories to be consistent with inertial frames and thus we work hard to write our equations and calculations in a lorentz invariant fashion.

Putting all this together, it has been fruitful over the last century to repeat quantum mechanics, perturbative and non-perturbative using the idea of fields - instead of single particle wavefunctions, while working mostly with Lorentz invariant equations for their dynamics. QFT produced some remarkable results over the journey, and we will discuss some of the classical topics in this article.

- ◇ **Scattering cross-sections** which are HEP's experimental core, are calculated upto a remarkable precision. We will indeed discuss in the end of this article, how QFT relates the abstract theory and computations to experimental predictions.
- ◇ QFT as a description of the fundamental interactions between elementary particles, when viewed from the naive perspective of *potentials*, does indeed recover the **Coulomb potential**! This will occupy us in the second article.
- ◇ One of the most accurate predictions of theoretical physics come from the QFT's calculation of the **anomalous magnetic moment** of electron. This occurs as a

loop corrections inside the realm of feynman diagrams and perturbation theory. Third article will communicate this from the perspective of **Renormalization**.



2. QUANTIZING FIELDS

2.1. Classical fields. Much of the Lagrangian mechanics extends to classical fields as dynamical variables. This includes, the notion of lagrangian (density), Noether's theorem, conserved quantities (currents), and symmetries.

2.2. Infinite Harmonic Oscillators. Expand the solution in terms of fourier modes. Discuss the algebra inheritance from harmonic oscillators. How is it connected and makes sense. Quantizing fields/hamiltonian, annihilation creation operators, Fock space...

2.3. Interactions and Cross-sections. Set the context for perturbation theory, and Feynman diagrams.

2.4. Feynman Diagrams. Give the rules for drawing diagrams.

Comment/give a hint as to how these rules are laid out.

2.5. Experiments in High Energy Physics. Connect to experiments as much as possible.

3. COLOUMB POTENTIAL À LA QFT

4. ANOMOLOUS MAGNETIC MOMENT OF ELECTRON